

JAKE GONZALES

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RESEARCH INTERESTS

My research lies at the intersection of control theory, machine learning, optimization, uncertainty quantification, and game theory, with a specific focus on decision-making in multi-agent systems. I am particularly interested in developing theoretical frameworks and principled algorithms that enable AI systems to reason about uncertainties, interact safely with other agents, and adapt effectively in dynamic, open-world environments.

EDUCATION

University of Washington <i>Ph.D., Electrical Engineering</i>	Sept. 2023 – Present Seattle, WA
• Advisors: Prof. Lillian Ratliff, Prof. Karen Leung, and Prof. Behçet Açıkmeşe	
University of New Mexico <i>Bachelor of Science in Electrical Engineering</i>	Aug. 2019 – May 2023 Albuquerque, NM
• Advisor: Prof. Meeko Oishi	

HONORS AND AWARDS

NSF Graduate Research Fellowship	2025
Invited to the first ever AI Startup School conference	2025
Amazon Ph.D. Fellowship, UW Amazon Science Hub [announcement]	2024 – 2025
Amazon Robotics Elevate Fellowship Funds (\$10,000 award)	2024
GEM Ph.D. Fellowship	2023
Department of Defense (DoD) Secret Security Clearance	2023
Department of Energy (DOE) Top Secret (Q) Security Clearance	2021
Hispanic Scholarship Fund (HSF) Scholar	2021, 2022
UNM Dean's List	2021, 2023

PUBLICATIONS

1. Max Horwitz, **Jake Gonzales**, Eric Mazumdar, Lillian J. Ratliff. “Structure from Strategic Interaction & Uncertainty: Risk Sensitive Games for Robust Preference Learning,” *under review*, 2026. [[Paper Link](#)]
2. **Jake Gonzales**, Max Horwitz, Eric Mazumdar, Lillian Ratliff. “Strategically Robust Multi-Agent Reinforcement Learning with Linear Function Approximation,” *Reinforcement Learning Conference (RLC)*, 2026. [[Paper Link](#)]
3. **Jake Gonzales**, Kazuki Mizuta, Karen Leung*, Lillian Ratliff*. “Safe Probabilistic Planning for Human-Robot Interaction using Conformal Risk Control,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. [[Project Website](#)]
4. Oswin So, Zachary Serlin, Makai Mann, **Jake Gonzales**, Kwesi Rutledge, Nicholas Roy, Chuchu Fan. “How to Train Your Neural Control Barrier Function: Learning Safety Filters for Complex Input-Constrained Systems,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. [[Paper Link](#)]
5. Adam J. Thorpe, **Jake A. Gonzales**, Meeko MK Oishi. “Data-Driven Stochastic Optimal Control Using Kernel Gradients,” *American Control Conference (ACC)*, 2023. [[Paper Link](#)]
6. Sofie W. Schunk, Shane McMurray, **Jake A. Gonzales**. “Advancing Model Credibility for Linked Multi-Physics Surrogate Models within a Coupled Digital Engineering Workflow of Nuclear Deterrence Systems,” *Model Validation and Uncertainty Quantification, Proceedings of the 41st IMAC*, 2023. [[Paper Link](#)]

7. Kelsey Wilson, Ruby Ta, **Jake Gonzales**, Seethamble S. Mani, Casey Noll, Wesley Krueger, William Gruner, Timothy Wisley. “Visualization of Model-based Systems Engineering Datasets in an Interactive 3D Game Engine,” *Western States Regional Conference INCOSE*, Sept. 2022.

RESEARCH EXPERIENCE

PhD Researcher Sept. 2023 – Present
University of Washington Seattle, WA

- Research in using game theory to improve robustness and generalization in vision-language-action (VLA) model fine-tuning for robotic manipulation.
- Research in risk-sensitive decision-making in multi-agent reinforcement learning using convex risk measures for robustness and generalization to out-of-distribution situations and unseen opponents in collaborative games.
- Developing methods for risk-sensitive preference learning for safe alignment of LLMs using game-theoretic techniques.
- Developing novel techniques for uncertainty quantification in embodied AI systems using conformal prediction and sequential testing frameworks.

Undergraduate Researcher Aug. 2021 – May 2023
University of New Mexico Albuquerque, NM

- Research in non-parametric methods for approximating solutions to stochastic optimal control problems using the theory of kernel embeddings of distributions resulting in efficient controller synthesis for nonlinear systems.
- Developed kernel gradient-based optimization algorithms for refining solutions to stochastic optimal control problems.

Undergraduate Researcher Aug. 2022
Stanford University Palo Alto, CA

- Short summer research program working with the Autonomous Systems Lab under Prof. Marco Pavone.
- Developed deep learning models for perception-based autonomous navigation through a hand-made driving course.

WORK EXPERIENCE

Amazon, Industrial Robotics Group June 2026 – Present
Robotics Applied Scientist II Intern Boston, MA

- Developing theory and algorithms for formal safety guarantees on dynamic, high-dimensional real-world robots operating under uncertainty and incorporating learning-based components such as reinforcement learning.

Amazon Robotics June 2025 – Sept. 2025
Applied Scientist II Intern Boston, MA

- Research and development in enabling uncertainty quantification for safe and reliable human perception systems.
- Developed generative models trained on real-world data for out-of-distribution detection and system robustness.
- Deployed models on real-world hardware systems, demonstrating capabilities to company leadership.

MIT Lincoln Laboratories June 2023 – Sept. 2023
Research Intern Boston, MA

- Developed algorithms for safe multi-agent control of nonlinear, high-dimensional systems with input constraints using neural control barrier functions.

Sandia National Laboratories March 2021 – July 2023
Undergrad Year-Round Intern Albuquerque, NM

- Worked on challenging problems related to the advancement of digital engineering for nuclear deterrence applications.
- Developed reduced-order multi-physics models of subcomponents of nuclear deterrence systems.
- Applied Sobol' sensitivity analysis to complex, nonlinear physical systems for uncertainty quantification.
- Built interactive VR environments that integrated various datasets for decision-makers to become experts on ND models.

TECHNICAL PRESENTATIONS

- Decision-Theoretic Uncertainty Adaptation in Multi-Agent Systems: Algorithms with Provable Guarantees
Qualifying Exam, Committee: Sam Burden (chair), Maryam Fazel, and Karen Leung, May 2025. [Slides]
- Hierarchical Framework for Scalable Multi-Agent Autonomous Mobility
Lightning Talk at ECE Research Showcase, University of Washington, March 2024. [Poster Link]
- Systems Engineering Leveraging a Commercial Gaming Platform
Western States Regional Conference INCOSE, Denver, CO, Sept. 2022.
- Fusing of Model-Based Systems Engineering and Virtual Reality
Sandia National Labs' 4th Annual XR Conference, virtual, July 2022.

TEACHING EXPERIENCE

- Teaching Assistant, Teaching Engineering** Spring 2024
University of Washington Seattle, WA
- Graded homework and class projects for 45+ students and gave a short lecture on effective presentation techniques.
- Teaching Assistant, Computer Logic Design** Spring, Fall 2022
University of New Mexico Albuquerque, NM
- Led lab sessions for 50+ students per semester and delivered short lectures on course topics.
- Teaching Assistant, Introduction to Electrical Engineering** Spring, Fall 2021
University of New Mexico Albuquerque, NM
- Led remote lab sessions and assisted 50+ students per semester with homework during COVID-19.

PROFESSIONAL SERVICE

- Reviewer, IEEE Robotics and Automation Letters (RA-L) 2026
- Reviewer, ICRA, IROS, CoRL (SAFE-ROL workshop) 2025

RELEVANT COURSEWORK

* indicates graduate-level courses

Control Theory: classical control theory, design of feedback control systems, linear systems theory*, linear multivariable control*, nonlinear control systems*

Mathematics: probability theory, advanced calculus*, convex optimization*, mathematical foundations of systems theory*, fundamental concepts in analysis*, advanced probability (measure-theoretic)*

Learning and Robotics: machine learning*, deep learning*, reinforcement learning*, autonomous mobile robots*, decision-making and control for safe interactive autonomy*, learning in games*

MENTORSHIP & OUTREACH

- Research Mentor**, University of Washington
- Max Horwitz, UW ECE & Pure Math undergraduate. Fall 2025 – Present
 - Aly Ahmed, UW CSE & Math undergraduate. Spring 2026 – Present
 - Alexandra Yankovskaya, UW ECE undergraduate. Spring 2026 – Present
- Research Mentor**, Tesla STEM High School Fall 2025 – Present
- Mentoring a high school student using control theory and AI for time-series prediction in financial markets.
- Graduate Student Mentor**, Machine Learning Reading Program (UW) Spring 2026
- Volunteered as a graduate student mentor in the UW MLRP, which pairs graduate and undergraduate students to read and discuss AI papers.

PhD Student Volunteer, University of Washington

Fall 2023, 2024, 2025

- Provided feedback to underrepresented prospective PhD students applying to UW ECE through a graduate application support program.

PhD Student Member, University of Washington

Fall 2024 – Spring 2025

- Served an academic term on the UW ECE DEI advisory committee to provide input on faculty/staff searches and advocate for initiatives promoting department inclusion.

Math Mentor, Prison Mathematics Project

2023 – 2025

- Mentored an inmate rehabilitating himself through mathematics.

Research Mentor, Tesla STEM High School

Fall 2023 – Spring 2024

- Mentored high school students using machine learning to model mercury pollution in aquatic ecosystems.

Peer Mentor, UNM Student Success Center

Fall 2022 – Spring 2023

- Mentored five new undergrad engineering students from traditionally underrepresented groups to provide help and guidance with challenges of being a first-year engineering student.

Chess Coach, Learners Chess Academy

2021 – 2023

- Taught chess at local K-8 schools in Albuquerque, NM to 100+ students.

TECHNICAL SKILLS

Programming: Python, C++, C

Scientific Computing: CVXPY, MATLAB/Simulink, NetworkX

ML Frameworks/Libraries: JAX, PyTorch, TensorFlow, scikit-learn

Software & Tools: MuJoCo MJX, mlab, ROS2, Unity, Gazebo, Docker, Arduino

REFERENCES

Prof. Lillian Ratliff

Associate Professor

Department of Electrical and Computer Engineering

University of Washington

Email: ratliff@uw.edu

Prof. Karen Leung

Assistant Professor

Department of Aeronautics and Astronautics

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Prof. Behçet Açıkmış

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